

C. Quotient and Remainder

time limit per test: 2 seconds

memory limit per test: 512 megabytes

You are given two integer arrays: $q_1, q_2, \dots, q_n$ and $r_1, r_2, \dots, r_n$, as well as an integer k .

You can perform the following operation any number of times (possibly zero):

- Choose two integers x and y such that
 - $1 \leq y < x \leq k$;
 - there exists an index i such that $q_i = \left\lfloor \frac{x}{y} \right\rfloor$ (rounded down);
 - there exists an index j such that $r_j = x \bmod y$.
- Remove q_i from the array q and r_j from the array r . If there are multiple occurrences of q_i in the array q , only one occurrence is removed; same for r_j and the array r .

Calculate the maximum number of operations that you can perform on the given arrays q and r .

Input

The first line contains one integer t ($1 \leq t \leq 10^4$) — the number of test cases.

The first line of each test case contains two integers n and k ($1 \leq n \leq 2 \cdot 10^5$; $2 \leq k \leq 10^{18}$) — the size of the arrays q and r and the upper limit for x and y .

The second line of each test case contains n integers $q_1, q_2, \dots, q_n$ ($1 \leq q_i \leq 10^9$) — the array q .

The third line contains n integers $r_1, r_2, \dots, r_n$ ($1 \leq r_i \leq 10^9$) — the array r .

Additional constraints on the input: the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, print one integer — the maximum number of operations that you can perform on the given arrays.

Example

input	Copy
<pre>3 1 100 1 27 3 10 5 6 5 7 1 7 5 42 5 4 2 2 1 9 8 9 8 100</pre>	
output	Copy
<pre>1 0 3</pre>	

Note

In the first test case, one operation can be performed: you can choose $x = 69$ and $y = 42$.

Then $\left\lfloor \frac{69}{42} \right\rfloor = 1 = q_1$ and $69 \bmod 42 = 27 = r_1$.

In the second test case, it is impossible to perform any operations, as suitable x and y cannot be found.

In the third test case, three operations can be performed:

- $x = 42, y = 17$: $\left\lfloor \frac{42}{17} \right\rfloor = 2 = q_3$ and $42 \bmod 17 = 8 = r_2$;

Educational Codeforces Round 185 (Rated for Div. 2)

Contest is running

00:51:07

Contestant



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Language: GNU G++17 7.3.0

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2. $x = 41, y = 16$: $\left\lfloor \frac{41}{16} \right\rfloor = 2 = q_4$ and $41 \bmod 16 = 9 = r_1$;
3. $x = 20, y = 12$: $\left\lfloor \frac{20}{12} \right\rfloor = 1 = q_5$ and $20 \bmod 12 = 8 = r_4$;

After these operations, you'll get arrays $q = [5, 4]$ and $r = [9, 100]$, and there are no more operations you can perform.

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